Process Scheduling and Deadlocks

The operating system kernel decides which process runs on the processor next. A scheduler may use round robin, priority or shortest job first scheduling for each process and thread.

A deadlock occurs when processes wait forever for resources held by each other. Programs use a semaphore or a mutex for synchronization, and careless locking of a mutex can cause deadlock.

A context switch saves the state of one process and loads another, usually after an interrupt from the timer.